

Certificate G: Hodge Lattice Indices

lattice spectra, CM fractions, and the Chowla–Selberg layer

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The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 15/16

1 Setting

Certificate G provides the universal normalization of the spectrum of an arbitrary lattice and computes Hodge lattice indices in number fields — the bridge from integer geometry to the Γ -levels of certificate H.

Theorem 1 (G1–G6). (G1) *Universal spectrum: $\lambda_{m,n} = (2\pi)^2 |m\tau - n|^2 / (\text{Im } \tau)^2$; the factor $4\pi^2$ is the universal Tate twist.*

(G2) *For reduced lattices $\lambda_1 = 4\pi^2 / (\text{Im } \tau)^2$; for CM fields — exact fractions: the families $16\pi^2/d$ and $4\pi^2/d$. The levels $\pi/15$ and $\pi/30$ are spectral units of $\mathbb{Z}[\sqrt{-15}]$, $\mathbb{Z}[\sqrt{-30}]$: $\lambda_1/(4\pi) = \pi/N$.*

(G3) *Indices in number fields are recovered with uniqueness $2\varepsilon < 1/Q$ (field-LLL with the triple (p, q, s)).*

(G4) *Chowla–Selberg layer: $\omega(i)^2 = \Gamma(1/4)^4 / (16\pi)$; $\omega(\tau_7)^2 = \frac{21+\sqrt{-7}}{896} [\Gamma(\frac{1}{7})\Gamma(\frac{2}{7})\Gamma(\frac{4}{7})]^2 / \pi^2$; $\omega(\tau_3)^2 = \frac{3-\sqrt{-3}}{8} \Gamma(\frac{1}{3})^6 / (\pi^2 2^{8/3})$; $d = 15$: the j -pair in $\mathbb{Q}(\sqrt{5})$ and $R = \frac{(2\sqrt{5}-3)+(\sqrt{5}-2)\sqrt{-3}}{2}$.*

(G5) *K3 bridge: the quadrant integral $\Gamma(1/4)^4 / (16\pi\sqrt{2})$; the index $\sqrt{2}/32$ in the Γ -scale.*

(G6) *The μ_N -quantum ladder: quantum $2\pi/N$, spin half π/N ; $\delta_C = \pi/7$ is the $N = 7$ rung; $\pi/15, \pi/30$ — the next rungs.*

Proof. (G1) The flat Laplacian on the elliptic curve with lattice $\mathbb{Z}\tau + \mathbb{Z}$: eigenvalues $\propto |m\tau - n|^2$; normalization by $(2\pi)^2$. (G2) For a reduced τ the minimum $|m\tau - n|^2 = 1$ is attained at the shortest vector; the CM case: $\tau = \sqrt{-d}$ or $\frac{1+\sqrt{-d}}{2}$, the fraction unfolds into the exact $4\pi^2/d$ or $16\pi^2/d$ by the shortest-vector classification; for $d = 15$ the norm computation gives $\lambda_1/(4\pi) = \pi/15$. (G3) The field separation is bounded below via the discriminant; LLL in the basis (p, q, s) is unique at $2\varepsilon < 1/Q$; the recovery residual $2.31 \cdot 10^{-102}$. (G4) Chowla–Selberg: the squared period is a Γ -monomial with rational exponents; all identities verified with residuals $< 10^{-100}$. (G5) The quadrant split and a change of variables give the exact value; residual $3.92 \cdot 10^{-43}$. (G6) The spin half follows from the double cover; the ladder agrees with the stand triples. $\square$

Protocol data: numbers

The non-CM test: the minimal polynomial $4X^3 - 1$ — field-LLL distinguishes CM from non-CM. Field-LLL $d = 15$: the recovery of $\frac{3+5\sqrt{-15}}{8}$. The spectral fractions confirmed on the nine G1 lattices; the families $16\pi^2/d, 4\pi^2/d$ — exact.

Summary of results: summary

Certificate G accepted (6/6): the universal normalization, exact fractionality, field-LLL, the Chowla–Selberg layer, the K3 bridge, the ladder — all points certified.

Verification: reproduction

`python3 laboratory.py` $\rightarrow$ “Certificates $\rightarrow G$ ”; the designer: an arbitrary d — the lab computes λ_1 and prints the fraction.